Wadim **Müller**

Senior Software Engineer · Hardware-Near Systems & Lab Automation

An den Mainwiesen 51, 63533 Mainhausen, Germany

(+49) 176-96526090 | wadim.mueller@mail.de | wafgo | wadim-mueller-b6b44b169

Summary

Senior software engineer with 15+ years of building software that drives real hardware: from Linux kernel drivers and board bring-up up to device integration, data pipelines and operator dashboards. At CMBlu I am the lead developer of the control platform for a fleet of redox-flow battery test stands. I defined its architecture from a blank page, integrate pumps, valves, sensors, power converters and lab cyclers over TCP/IP, Modbus, REST, ZeroMQ and MQTT, and own the path from raw measurement to time-series storage and the operator's screen. Earlier: firmware for oscilloscopes, spectrum analysers and power meters at Rohde & Schwarz and safety-relevant multicore systems at Continental. Beyond engineering, I bring hands-on computational chemistry (DFT, semi-empirical methods), so I am at home working at the boundary to physicists, chemists and hardware engineers, and taking ownership of quality, testability and long-term reliability.

Experience

CMBlu Energy AG

Alzenau, Germany

Embedded Linux Developer · Lead Developer Battery Control Platform — *since Feb. 2025*

- **Greenfield architecture:** defined and built the control platform (Rust/Tokio) that runs battery management and process control on every redox-flow test stand: a component framework decoupled through a reactive datastore, configured per platform in YAML, with 50+ component types across ten hardware variants.
- **Device and instrument integration:** BioLogic battery cyclers (REST API, ZeroMQ event stream, WebSockets), TRUMPF AC/DC and DC/DC power converters and ICP DAS I/O modules (Modbus TCP), Mettler Toledo M800 transmitters, flow, pressure and temperature sensors (I^2C, UART, IIO), and an RPMsg bridge to safety firmware on a Cortex-M4 core.
- **Control and safety:** closed-loop PI flow control of pumps and valves, a rule-based safety engine with debounced threshold alarms and a latched software E-STOP, and interlocks that keep deployments and updates from interrupting a running experiment.
- **Operator HMIs and dashboards:** SCADA screens (FUXA) with animated process diagrams, live values and operator controls; fleet dashboards; and tooling that turns HMI projects into reviewable, automatically validated code.
- **Data pipelines:** telemetry over MQTT/Protobuf into InfluxDB and Azure Event Hub, journal logs into Loki, and a read-only observability service that correlates logs, time series and cycler state across a fleet of 30+ nodes to diagnose anomalies.
- **Engineering practice:** set up the team's quality gates: CI with linting and static analysis, mandatory unit-test coverage per component, tests traceable to requirements, merge-request reviews, and a QEMU-based emulation of the target (incl. a mainline-submitted AM64x SoC model) for testing without hardware.
- Own the Yocto-based Linux distribution for the TI AM64x platform: board bring-up, BSP, kernel drivers (counter, PWM, GPIO, I^2C sensors; one upstreamed to mainline), RPM feed and a guarded over-the-air update path.
- **Quantum chemistry / DFT:** built the high-throughput virtual-screening service chemists use to rank redox-active electrolyte candidates before synthesis. Computational workflow from SMILES to redox properties: conformer search (CREST), semi-empirical GFN2-xTB geometry optimisation and Boltzmann weighting, then DFT single points with Psi4 (hybrid and range-separated functionals such as B3LYP-D3(BJ) and ωB97X-D, def2 basis sets). Software side: web application with 3D molecule visualisation, a compute back end scheduling jobs against a CPU and memory budget, and a REST API on top of a Spark/Databricks pipeline.
- **ML soft-sensor:** replaced ORP probes that exist only on test rigs with a per-side state-of-charge model from operating signals alone (LightGBM, 47 engineered features, held-out MAE 8%, R^2 0.75). Assembled the 2.5M-row training corpus from InfluxDB time series, Grafana operator annotations, test-order sheets and the lab database; cleaned swapped, stuck and implausible sensor channels; calibrated inline sensors against lab reference samples; wrote a pure-Rust inference engine that runs the model on the embedded target.

Continental Automotive GmbH

Babenhausen, Germany

Software System Integrator · Embedded Linux Developer — *Jul. 2018 - Jan. 2025*

- Integrated vendor Board Support Packages into instrument-cluster products on Renesas multicore ARMv7/v8 SoCs running Green Hills INTEGRITY.
- Designed and deployed the CMake-based build systems of several instrument-cluster products.
- Debugged system-level failures on multicore ARMv8/v7 systems via JTAG and trace (Green Hills MULTI, Lauterbach Trace32, ARM ETM), and built in-house tooling for it (offline stack unwinding, ETM setup).
- Built QEMU machine emulation of customised Renesas R-Car3 boards so that software could be developed and tested before hardware was available (github.com/wafgo/qemu).

Magna Powertrain

Sailauf, Germany

Embedded Software Engineer — *Jan. 2017 - Jul. 2018*

- Implemented brushless DC motor-control algorithms on resource-limited microcontrollers and ported FreeRTOS to the NXP S12Z.
- Designed an MCU-agnostic software architecture for drive-control algorithms, with a kbuild-style build system on GNU make.

Rohde & Schwarz

Mainhausen, Germany

Embedded Software Engineer

Jan. 2009 - Dec. 2016

- **Instrument firmware:** developed embedded firmware for test and measurement instruments: oscilloscopes, spectrum analysers, power meters, LCR meters and multimeters, from board bring-up to measurement application modules with unit tests.
- Implemented the USB TMC remote-control interface shared by several instrument families.
- Wrote Linux kernel drivers and bring-up software for custom i.MX6 and Renesas SH2A boards, incl. an SPI-to-Ethernet bridge (PIC32 firmware plus Linux network driver).

Skills

Programming	C, C++, Rust, Python, Bash, Perl
Device Interfaces	TCP/IP, REST, WebSockets, MQTT, Protobuf, ZeroMQ, Modbus TCP, USB TMC, RPMsg, I^2C, SPI
Data & Monitoring	InfluxDB, Grafana, Loki, Prometheus, Azure Event Hub, PySpark/Databricks
HMI & Vision	SCADA HMIs (FUXA), web dashboards, OpenCV, V4L2, stereo-camera calibration
Systems & Low-Level	Linux kernel drivers, Yocto, U-Boot, FreeRTOS, QEMU/KVM, Docker
Quality & Debugging	GitLab CI, automated testing, code review, GDB, Trace32, JTAG/ETM
Quantum Chemistry	DFT (Psi4), semi-empirical GFN2-xTB, CREST, DFTB+, RDKit
ML & AI	LightGBM, PPO, PyTorch, Kalman filters, agentic coding, MCP servers, Ollama
Languages	German, English, Russian

Selected Projects

Spinoza – Table-Tennis Return Robot

github.com/wafgo/spinoza

Physics simulation, ML and closed-loop robotics

2026

- **Physics engine in Rust:** RK4 at 0.5 ms with drag, Magnus force and a stick/slip bounce model; PyO3 bindings feed a PPO agent (Stable-Baselines3) and a 1D-CNN trajectory predictor.
- **Hardware loop at 30 Hz:** calibrated stereo camera, triangulation and a constant-acceleration Kalman filter drive analytical IK for a 4-DOF servo arm.
- Built agent-first with AI coding assistants (Claude, Copilot), the workflow I also use in production.

Open Source

Linux Kernel (mainline)

lore.kernel.org

Maintainer and Contributor

since 2022

- Author and maintainer of the upstream IIO driver for the Sensirion SLF3S liquid-flow sensor (v7.3), incl. devicetree binding, IIO core additions and KUnit tests.
- Initial mainline bring-up of the NXP S32G3 SoC; RISC-V support for `checkstack.pl`; a generic GPIO-based counter driver and a PCI-endpoint block-device passthrough (covered on LWN), both under review.

QEMU, Psi4, thin-edge.io and others

github.com/wafgo

Contributor

- **QEMU:** TI AM64x SoC model and machine that runs the stock TI boot chain end to end (ROM, R5 SPL, TF-A/OP-TEE, U-Boot, SMP Linux); fix and qtest for an SDHCI ADMA bug.
- **Psi4 (quantum chemistry):** GauXC backend for the DFT exchange-correlation quadrature with optional GPU offload (∼1k lines C++), after profiling showed it at 70 % of SCF time; superseded by a parallel upstream effort.
- **thin-edge.io (Rust):** MQTT connection-leak fix (merged). Further: meta-tedge, pyelftools, GEF, Tinkerforge, radare2, pahole.

Education

Technische Hochschule Aschaffenburg (University of Applied Sciences)

Aschaffenburg, Germany

B.Eng. in Electrical Engineering and Information Technology

Sep. 2008 - Aug. 2012

- Microelectronics, final grade 1.5 (First Class Honours). Thesis: USB TMC device interface.

Hochschule Darmstadt (University of Applied Sciences)

Darmstadt, Germany

M.Sc. in Electrical Engineering

Sep. 2014 - Aug. 2017

- Part-time study, Microelectronics, final grade 1.1 (First Class Honours). Thesis: FPGA-accelerated Linux system for real-time depth mapping from a calibrated stereo camera (V4L2, OpenCV): HLS IP cores on a Zynq SoC, UIO driver, application (github.com/wafgo/rt-depth-map).